Sheet 79 Thu. 10 June 2027 Presentation of the specialist subjects; review

1. Warm-up · Flash Express $\frac{3}{8}$ as a percentage.
2. Core · Metacognition What are your three revision priorities before Première?
3. Core + · Flash Solve 2 $x^2 = 18$.
4. Back · Trap Calculate $(-2)^4$ and -2^4.
5. Running thread · True/false True or false? Justify. If an integer is divisible by 4 and by 6, then it is divisible by 24.

Annual review and consolidation of Seconde learning

Sheet 80 Mon. 14 June 2027 Annual review and consolidation

1. Warm-up · Flash $A(-1;4)$ and $B(3;1)$. Coordinates of $\overrightarrow{AB}$?
2. Core · Method Slope-intercept equation of (AB) with $A(0;1)$ and $B(2;5)$.
3. Core + · Method $A(1;0)$, $B(4;1)$, $C(5;4)$. Determine D so that $ABCD$ is a parallelogram.
4. Back · Flash Calculate $-5+8\times(-2)$.
5. Running thread · Statement→ formula The area of a square with side x is a function A of x. Give $A(x)$, a table of values for $x = 1; 2; 3$, and say whether A is proportional to x.

Sheet 81 Tue. 15 June 2027 Annual review and consolidation

1. Warm-up · Flash Write 0.00045 in standard form.
2. Core · Flash Determine $[-2;3[\cap]0;5]$.
3. Core + · Method Solve $|x-1|<4$ and give S as an interval.
4. Back · Expand Expand and simplify $(2x-1)^2$.
5. Running thread · Flash A cylinder has radius 3 cm and height 5 cm. Calculate its exact volume.

Sheet 82 Thu. 17 June 2027 Annual review and consolidation

1. Warm-up · Flash $f(x) = x^2 - 4x$. Calculate $f(3)$.
2. Core · Flash Solve $x^2 = 9$ then $|x| = 2$.
3. Core + · Method $f(x) = 2x - 3$ and $g(x) = x + 1$. Solve $f(x) < g(x)$.
4. Back · Flash A tap delivers 18 L in 4 min. What volume in 10 min?
5. Running thread · Reading code What is the final value of `n`?

```
n = 0
while n*n < 30:
    n = n + 1
```

Sheet 83 Mon. 21 June 2027 Annual review and consolidation

1. Warm-up · Flash Mean of 5, 7, 9, 11?
2. Core · Method Data set 1; 3; 4; 6; 8; 9; 11: Q_1, median, Q_3.
3. Core + · True/false True or false? Justify. Adding 10 to each value of a data set leaves its standard deviation unchanged.
4. Back · Reflection How many preimages do -1 and 4 have for $f(x) = x^2$?
5. Running thread · Prove Show that for every real number x: $(x-1)(x+1)+1 = x^2$.

Sheet 84 Tue. 22 June 2027 Annual review and consolidation

1. Warm-up · Flash Expand and simplify $(2x-3)(x+4)$.
2. Core · Flash Solve 3 $x-7 \leqslant 2x+5$.
3. Core + · Method Solve $(x-1)(x+4)=0$ then $x^2=5$.
4. Back · Reflection Subscription: 5 € + 3 € per entry. Is the price proportional to the number of entries?
5. Running thread · Flash A cone has radius 3 cm and height 4 cm. Calculate its slant height and its exact volume.

Sheet 85 Thu. 24 June 2027 Annual review and consolidation

1. Warm-up · Flash What multiplier corresponds to an 8% increase?
2. Core · Method A 25% increase is followed by a 20% decrease. What is the overall rate?
3. Core + · Flash A price goes from 5,000 to 5,750 Ar. Rate of change?
4. Back · Recognise What is the cross-section of a cylinder by a plane containing its axis? By a plane perpendicular to its axis?
5. Running thread · Problem Three times a number, minus 4, equals twice that number plus 5. Write an equation then find the number.

Sheet 86 Mon. 28 June 2027 Annual review and consolidation

1. Warm-up · Flash Direction of variation of $x \mapsto -3x+2$?
2. Core · Method Minimum and maximum of $f(x)=x^2$ on $[-3;1]$?
3. Core + · Method For $x>0$, compare $\frac{1}{x}$ and $\frac{1}{x+1}$.
4. Back · Construction steps Describe in three steps how to construct the image A' of A under reflection in a line (d).
5. Running thread · Method A point I satisfies $IA=IB$. Where is it? Which property?

Sheet 87 Tue. 29 June 2027 Annual review and consolidation

1. Warm-up · Flash $P(A)=0.28$. Calculate $P(\overline{A})$.
2. Core · Method 100 people including 60 women; 6 women and 4 men are left-handed. Calculate $P_F(G)$ and $P_G(F)$.
3. Core + · Method $P(A)=0{,}4$, $P_A(B)=0{,}5$, $P_{\overline{A}}(B)=0.2$. Calculate $P(B)$.
4. Back · Problem Volume of a cube with edge x : $V(x)=x^3$. Calculate $V(3)$ and find x such that $V(x)=64$.
5. Running thread · Flash In which quadrant (numbered I to IV anticlockwise) is the point $M(a;b)$ with $a<0$ and $b>0$?

Sheet 88 Thu. 1 st July 2027 Annual review and consolidation

1. Warm-up · Flash Hypotenuse of a right-angled triangle with shorter sides 5 and 12?
2. Core · Flash Midpoint of $[AB]$ with $A(-3;2)$ and $B(5;-4)$?
3. Core + · Write up Show that $A(0;0)$, $B(2;1)$, $C(-1;2)$ form a triangle that is right-angled and isosceles at A.
4. Back · Flash Give an approximate value of $\frac{1}{3}$ à 10^{-2} accuracy. Is this number a decimal?
5. Running thread · Problem What is the smallest integer that leaves remainder 1 when divided by 4 and by 6, greater than 1?